Solutions to Aluffi's Algebra: Chapter 0

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V: Irreducibility and factorization in integral domains

§1 Chain conditions and existence of factorizations

1.1

Let R be a Noetherian ring, and let I be an ideal of R. Prove that R/I is a Noetherian ring.

Solution.
Suppose J/I is an ideal of R/I, corresponding to an ideal J of R, $J \supset I$. Since R is Noetherian, $J = (a_1, \ldots, a_n) \implies J/I = (\overline{a}_1, \ldots, \overline{a}_n)$. $\square$

1.2

Prove that if $R[x]$ is Noetherian, so is R. (This is a 'converse' to Hilbert's basis theorem.)

Solution.
$R \simeq \frac{R[x]}{(x)}$, hence R is Noetherian by the previous exercise. $\square$

1.3

Let k be a field, and let $f \in k[x]$, $f \notin k$. For every subring R of $k[x]$ containing k and f, define a homomorphism $\varphi : k[t] \to R$ by extending the identity on k and mapping t to f. This makes every such R a $k[t]$-algebra.

- Prove that $k[x]$ is finitely generated as a $k[t]$-module.
- Prove that every subring R as above is finitely generated as a $k[t]$-module.

- Prove that every subring of $k[x]$ containing k is a Noetherian ring.

Solution.
Let $\deg f = n$ and let us show that every $g \in k[x]$ can be expressed as

$$g(x) = \sum a_i(x) f(x)^i$$

with $\deg a_i < n$.

The statement trivially holds for all g such that $\deg g < n$. If $\deg g \geq n$, then we can write $g = qf + r$, where $\deg q < \deg g$ and $\deg r < n$. We may then inductively assume that $q = \sum a_i f^i$, with $\deg a_i < n$, which implies

$$g = \left(\sum a_i f^i\right) f + r = \sum a_i f^{i+1} + r = \sum a_i^* f^i$$

It follows then that for all $g \in k[x]$:

$$g(x) = \sum a_i(x) f(x)^i = \sum_{i=0}^{m} \sum_{j=0}^{n-1} a_{ij} x^j f(x)^i = \sum_j x^j \sum_i a_{ij} f(x)^i = \sum p_j(t) \cdot x^j$$

where $p_j(t) = \sum_i a_{ij} t^i$.

Thus, $k[x] = \langle 1, x, \ldots, x^{n-1} \rangle$ as a $k[t]$-module.

If R is a subring of $k[x]$ containing k and f, then R is a submodule of $k[x]$ over $k[t]$. Since $k[t]$ is a PID and $k[x]$ is finitely generated, R is also finitely generated.

If R is a subring of $k[x]$ containing k, then either $R = k$, in which case R is Noetherian, or there exists $f \in R$ such that $f \notin k$, which defines a $k[t]$-module structure on $k[x]$, making R into a finitely generated submodule.

Suppose I is an ideal of R. I is a submodule of $k[x]$. Indeed, it is closed under addition and $\forall g \in I \; \forall p \in k[t]$,

$$p(t) \cdot g = \left(\sum a_i f(x)^i\right) \cdot g(x) \in I,$$

since $\sum a_i f^i \in R$.

Therefore, I is finitely generated as a $k[t]$-module. Let $g_1, \ldots, g_m$ be its generators. It follows that $I = (g_1, \ldots, g_m)$ as an ideal of R. □

1.4

Let R be the ring of real-valued continuous functions on the interval $[0, 1]$. Prove that R is not Noetherian.

Solution.
Define $N_A = \{f \in R \mid f(a) = 0 \; \forall a \in A\}$. For any nonempty $A \subset [0, 1]$, N_A is clearly an ideal.

The chain $N_{[0,1]} \subset N_{[0,\frac{1}{2}]} \subset \dots \subset N_{[0,\frac{1}{n}]} \subset \dots$ does not stabilize, hence R is not Noetherian. $\square$

1.8

Prove that every ideal in a Noetherian ring R contains a finite product of prime ideals. (Hint: Let $\mathcal{F}$ be the family of ideals that do not contain finite products of prime ideals. If $\mathcal{F}$ is nonempty, it has a maximal element M since R is Noetherian. Since $M \in \mathcal{F}$, M is not itself prime, so $\exists a, b \in R$ s.t. $a \notin M, b \notin M$, yet $ab \in M$. What's wrong with this?)

Solution.
Since $M + (a)$ and $M + (b)$ strictly contain M, both ideals are not in $\mathcal{F}$. Hence, there are prime ideals $P_1, \dots, P_n, P'_1, \dots, P'_m$ such that $P_1 \dots P_n \subset M + (a)$ and $P'_1 \dots P'_m \subset M + (b) \implies P_1 \dots P_n P'_1 \dots P'_m \subset (M + (a))(M + (b)) \subset M + (ab) = M$. This contradicts the fact that M does not contain a finite product of prime ideals. $\square$

1.9

Let R be a commutative ring, and let $I \subsetneq R$ be a proper ideal. The reader will prove in Exercise 3.12 that the set of prime ideals containing I has minimal elements (the *minimal primes* of I). Prove that if R is Noetherian, then the set of minimal primes of I is finite. (Hint: Let $\mathcal{F}$ be the family of ideals that do *not* have finitely many minimal primes. If $\mathcal{F} \neq \emptyset$, note that $\mathcal{F}$ must have a maximal element I, and I is not prime itself. Find ideals J_1, J_2 strictly larger than I, such that $J_1 J_2 \subseteq I$, and deduce a contradiction.)

Solution.
I claim that if $I \subset J \subset P$ and P is a minimal prime of I, then P is a minimal prime of J. Indeed, if there is a prime ideal P' such that $J \subset P' \subsetneq P$, then $I \subset P' \subsetneq P$, which contradicts the minimality of P with respect to I.

Now if I is a maximal element of $\mathcal{F}$ defined as above, then I is not itself prime, for otherwise the set of minimal primes of I would consist of a single element: I. Hence, $\exists a, b \in R$ such that $a \notin I, b \notin I, ab \in I \implies I + (a) \supsetneq I$, $I + (b) \supsetneq I \implies I + (a)$ and $I + (b)$ have finite sets of minimal primes. Let us denote them by M_a and M_b. We also have $(I + (a))(I + (b)) \subset I + (ab) = I$ and thus, if P is a minimal prime of I, then $(I + (a))(I + (b)) \subset P$, and therefore by exercise III.4.19 $I + (a) \subset P$ or $I + (b) \subset P$, which as discussed before implies that P is a minimal prime of $I + (a)$ or $I + (b)$. This gives us a mapping from the collection M of minimal primes of I into $M_a \cup M_b$, and this mapping is obviously injective. It is impossible, since M is infinite and $M_a \cup M_b$ isn't. $\square$

1.10

By Proposition 1.1, a ring R is Noetherian if and only if it satisfies the a.c.c. for ideals. A ring is *Artinian* if it satisfies the d.c.c. (descending chain condition) for ideals. Prove that if R is Artinian and $I \subseteq R$ is an ideal, then R/I is Artinian. Prove that if R is an Artinian integral domain, then it is a field. (Hint: Let $r \in R$, $r \neq 0$. The ideals (r^n) form a descending sequence; hence $(r^n) = (r^{n+1})$ for some n. Therefore...) Prove that Artinian rings have Krull dimension 0 (that is, prime ideals are maximal in Artinian rings).

Solution.
Suppose R is Artinian, $I \subset R$ is an ideal and $J_1 \supset J_2 \supset \dots$ is a descending chain of ideals in R/I. Recall that there is a natural inclusion-preserving correspondence between ideals of R/I and ideals of R containing I. Therefore, there is a chain $I_1 \supset I_2 \supset \dots$ in R, which must stabilize because R is Artinian. It follows that $J_1 \supset J_2 \supset \dots$ also stabilizes.

Now if R is an Artinian integral domain, then $\forall r \in R, r \neq 0, \exists n \in \mathbb{N}$ such that $(r^n) = (r^{n+1}) \implies \exists a \in R$ such that $r^n = ar^{n+1} \implies r^n(1 - ar) = 0$. Since R is an integral domain and $r \neq 0$, we have $1 - ar = 0$, so r is invertible.

Finally, we need to show that prime ideals are maximal if R is Artinian. Assume $P \subset R$ is prime. Then, as shown before, R/P is also Artinian. Moreover, it is an integral domain because P is prime, and hence it is a field $\implies$ P is maximal. □

1.13

Prove that, for nonzero elements, prime $\iff$ irreducible in $\mathbb{Z}$.

Solution.
Primes are irreducibles in all integral domains, so we only need to show sufficiency. If $p \neq 0$ is irreducible in $\mathbb{Z}$, then (p) is maximal among principal ideals of $\mathbb{Z}$. Since in $\mathbb{Z}$ every ideal is principal, (p) is maximal among all ideals of $\mathbb{Z}$, hence (p) is prime and p is prime by definition. □

1.14

For a, b in a commutative ring R, prove that the class of a in $R/(b)$ is prime if and only if the class of b in $R/(a)$ is prime.

Solution.
Let us denote by $[(a)]_{(b)}$ the image of $(a) \subset R$ under the canonical projection on $R/(b)$. Using the result in III.3.3 we get:

$$\overline{a} \text{ is prime in } R/(b) \iff (\overline{a}) = [(a)]_{(b)} \text{ is prime in } R/(b)$$

$$\iff \frac{R/(b)}{[(a)]_{(b)}} \simeq \frac{R}{(a)+(b)} \simeq \frac{R/(a)}{[(b)]_{(a)}} \text{ is an integral domain}$$
$$\iff [(b)]_{(a)} = (\bar{b}) \text{ is prime in } R/(a) \iff \bar{b} \text{ is prime in } R/(a).$$

□

1.17

Consider the subring of $\mathbb{C}$:
$$\mathbb{Z}[\sqrt{-5}] := \{a + bi\sqrt{5} \mid a, b \in \mathbb{Z}\}.$$

- Prove that this ring is isomorphic to $\mathbb{Z}[t]/(t^2+5)$.
- Prove that it is a Noetherian integral domain.
- Define a 'norm' N on $\mathbb{Z}[\sqrt{-5}]$ by setting $N(a+bi\sqrt{5}) = a^2+5b^2$. Prove that $N(zw) = N(z)N(w)$. (Cf. Exercise III.4.10.)
- Prove that the units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1. (Use the preceding point.)
- Prove that 2, 3, $1+i\sqrt{5}$, $1-i\sqrt{5}$ are all irreducible nonassociate elements of $\mathbb{Z}[\sqrt{-5}]$.
- Prove that no element listed in the preceding point is prime. (Prove that the rings obtained by mod-ing out the ideals generated by these elements are not integral domains.)
- Prove that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Solution.
$\varphi : \mathbb{Z}[\sqrt{-5}] \to \frac{\mathbb{Z}[t]}{(t^2+5)}$ defined by $\varphi(a+bi\sqrt{5}) = \overline{a+bt}$ is an isomorphism, which can be trivially verified. Let's just check that φ preserves products:
$$\varphi((a+bi\sqrt{5})(a'+b'i\sqrt{5})) = \varphi((aa'-5bb') + (ab'+a'b)i\sqrt{5}) =$$
$$= \overline{(aa'-5bb') + (ab'+a'b)t} = (\text{since } \bar{t}^2 = \overline{-5}) =$$
$$= \overline{(a+bt)} \cdot \overline{(a'+b't)} = \varphi(a+bi\sqrt{5})\varphi(a'+b'i\sqrt{5})$$
$\mathbb{Z}[\sqrt{-5}]$ is an integral domain because $\mathbb{C}$ is and it is Noetherian because $\mathbb{Z}[t]$ is.

We can write N as
$$N(a+bi\sqrt{5}) = a^2+5b^2 = (a-bi\sqrt{5})(a+bi\sqrt{5}) = (a+bi\sqrt{5}) \cdot \overline{(a+bi\sqrt{5})},$$
where $\overline{x}$ denotes a complex conjugate, so $N(x) = x\overline{x}$, which implies
$$N(zw) = zw\overline{zw} = z\overline{z}w\overline{w} = N(z)N(w)$$
Note that $N(z) \in \mathbb{Z}_{\geq 0}\ \forall z \in \mathbb{Z}[\sqrt{-5}]$.

If $u \in \mathbb{Z}[\sqrt{-5}]$ is a unit, then
$$1 = N(1) = N(uu^{-1}) = N(u)N(u^{-1}) \implies N(u) = 1$$
If $a^2+5b^2 = 1$ with $a, b \in \mathbb{Z}$, then $(a,b) \in \{(1,0), (-1,0)\}$ by size consideration. So the only units are 1 and -1.

The following trivial claims will help us to deal with irreducibility:

1. $\nexists v \in \mathbb{Z}[\sqrt{-5}] \mid N(v) = 2$ or $N(v) = 3$
2. If $v \mid w$, then $N(v) \mid N(w)$
3. If $N(v) = 1$ then v is a unit

To show that 2 is irreducible, assume $2 = vw \implies N(v) \mid N(2) = 4$. Since $N(v) \neq 2$, we must have $N(v) = 1$ or 4, which forces v or w to be a unit, proving that 2 is irreducible.

The same reasoning yields that 3 and $1 \pm i\sqrt{5}$ are irreducible.

None of the considered elements are prime, because $6 = 2 \cdot 3 = (1 - i\sqrt{5})(1 + i\sqrt{5})$. For example, $1 - i\sqrt{5} \mid 2 \cdot 3$, yet $1 - i\sqrt{5} \nmid 2$ and $1 - i\sqrt{5} \nmid 3$, since $N(1 - i\sqrt{5}) = 6 \nmid 4 = N(2)$ and $6 \nmid 9$.

The same factorization $6 = 2 \cdot 3 = (1 - i\sqrt{5})(1 + i\sqrt{5})$ also shows that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD. $\square$

§2 UFDs, PIDs, Euclidean domains

2.3

Let n be a positive integer. Prove that there is a one-to-one correspondence preserving multiplicities between the irreducible factors of n (as an integer) and the composition factors of $\mathbb{Z}/n\mathbb{Z}$ (as a group). (In fact, the Jordan-Hölder theorem may be used to prove that $\mathbb{Z}$ is a UFD.)

Solution.
Suppose $\mathbb{Z}_n = G_0 \supset G_1 \supset \cdots \supset G_r = 0$ is a composition series for $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$. If for some i, $|G_i/G_{i+1}| = p$ is reducible in $\mathbb{Z}$, then $\exists a, b \in \mathbb{Z} : |a| > 1, |b| > 1, p = ab$. Since $G_i = \langle \overline{d} \rangle$ and $G_{i+1} = \langle \overline{d'} \rangle$ for some $d, d' \in \mathbb{Z}$ s.t. $d \mid n$, $d' \mid n$, we have

$$ab = p = |G_i/G_{i+1}| = |G_i|/|G_{i+1}| = \frac{n/d}{n/d'} = \frac{d'}{d}$$

Thus, $G_i = \langle \overline{d} \rangle \supsetneq \langle \overline{ad} \rangle \supsetneq \langle \overline{abd} \rangle = \langle \overline{d'} \rangle = G_{i+1} \implies \implies G_i/G_{i+1}$ is not simple, which is a contradiction.

Note that $n = |\mathbb{Z}_n| = |G_i||\mathbb{Z}_n/G_i| = |G_{i+1}||G_i/G_{i+1}||\mathbb{Z}_n/G_i|$, so $|G_i/G_{i+1}| \mid n$. Hence, we conclude that the orders of all composition factors of $\mathbb{Z}_n$ are irreducible factors of n in $\mathbb{Z}$.

Moreover, if p is an irreducible factor of n of order k, so that $p^k \mid n$, $p^{k+1} \nmid n$, then there are exactly k composition factors of $\mathbb{Z}_n$ of order p. To see this, note that the composition series of $\mathbb{Z}_n$ identifies a sequence of integers $d_i = |G_i|$ s.t. $n = d_0 \vdots d_1 \vdots \ldots \vdots d_r = 1$. Since $p^k \mid d_0$ and $p \nmid d_r$, for each $s = \overline{1, k}$ there must exist an index i_s s.t. $p^s \mid |G_{i_s}|$ and $p^s \nmid |G_{i_s+1}|$. We have then $|G_{i_s}| = |G_{i_s+1}| \cdot |G_{i_s}/G_{i_s+1}| \implies p \mid |G_{i_s}/G_{i_s+1}|$, which forces $|G_{i_s}/G_{i_s+1}| = p$.

i_s are clearly not equal for different s, the desired statement follows. □

2.6

Let R be a domain with the property that the intersection of any family of principal ideals in R is necessarily a principal ideal.

- Show that greatest common divisors exist in R.
- Show that UFDs satisfy this property.

Solution.
If $a, b \in R$, then $(a) \cap (b) = (l)$ for some $l \in R$. $ab \in (a) \cap (b) \implies l \mid ab \implies ab = ld$ for some $d \in R$. I claim that d is gcd of a and b. Indeed, $l \in (a) \cap (b) \implies l \in (a) \implies l = l_a a \implies ab = l_a ad \implies b = l_a d \implies d \mid b$. $l \in (b)$ similarly yields $d \mid a$. If $d' \in R$ s.t. $d' \mid a$, $d' \mid b$, then

$a = a'd', b = b'd' \implies d'a'b' \in (a) \cap (b) = (l) \implies \implies d'a'b' = lm \implies ld = ab = (a'd')(b'd') = d'(d'a'b') = d'lm \implies d = d'm \implies d' \mid d$.

Suppose now R is a UFD and let $\mathcal{F} = \{(a_i)\}_{i\in I}$ be a family of principal ideals in R. We wish to show $\bigcap_{i\in I}(a_i) = (d)$. If $\bigcap_{i\in I}(a_i) = (0)$, it is trivially principal. Assume the intersection is non-zero, and let $0 \neq x \in \bigcap_{i\in I}(a_i)$.

Since x is in the intersection, $a_i \mid x$ for all $i \in I$. In a UFD, x has a finite number of irreducible factors up to associates, say $p_1, \dots, p_k$. Because $a_i \mid x$, every a_i must factor exclusively into these p_j. Let $n_j(a_i)$ be the exponent of p_j in the factorization of a_i. Since $a_i \mid x$, these exponents are strictly bounded: $n_j(a_i) \leq n_j(x)$ for all $i \in I$.

Thus, for each $j \in \overline{1,k}$, we can take the finite maximum exponent across the entire family: $M_j = \max_{i\in I} n_j(a_i)$. Let $d = p_1^{M_1} \dots p_k^{M_k}$.

Since $n_j(a_i) \leq M_j$ for all i, we have $a_i \mid d$, meaning $d \in \bigcap_{i\in I}(a_i)$, so $(d) \subseteq \bigcap_{i\in I}(a_i)$. Conversely, if $y \in \bigcap_{i\in I}(a_i)$, then $a_i \mid y$ for all i. Thus $n_j(a_i) \leq n_j(y)$ for all i, which forces $M_j \leq n_j(y)$. Therefore, $d \mid y$, meaning $y \in (d)$, so $\bigcap_{i\in I}(a_i) \subseteq (d)$.

Hence, $\bigcap_{i\in I}(a_i) = (d)$. □

2.7

Let R be a Noetherian domain, and assume that for all nonzero a, b in R, the greatest common divisors of a and b are linear combinations of a and b. Prove that R is a PID.

Solution.
Let us prove that

$$\gcd(a_1, \dots, a_{n+1}) = \gcd(\gcd(a_1, \dots, a_n), a_{n+1})$$

Denote $\gcd(a_1, \dots, a_n) = d$ and $\gcd(a_1, \dots, a_{n+1}) = d'$. We have $d' \mid a_i \quad \forall i = \overline{1, n+1} \implies d' \mid d$ and $d' \mid a_{n+1}$. If d^* is such that $d^* \mid d$ and $d^* \mid a_{n+1}$, then $d^* \mid a_i \quad \forall i = \overline{1, n+1} \implies d^* \mid d'$, so $d' = \gcd(d, a_{n+1})$.

It follows inductively that $\gcd(a_1, \dots, a_n) = d$ is a linear combination of $a_1, \dots, a_n$, hence if $I = (a_1, \dots, a_n)$ is an ideal, then $d = \sum r_i a_i \in I \implies (d) \subset I$. On the other hand, $I \subset (d)$ since all a_i are divisible by $d \implies I = (d)$. □

2.8

Let R be a UFD, and let $I \neq (0)$ be an ideal of R. Prove that every descending chain of principal ideals containing I must stabilize.

Solution.
Suppose $(a_1) \supset (a_2) \supset \dots$ is a descending chain of principal ideals s.t. $\forall k, (a_k) \supset I$. Let i be any non-zero element of I. $\forall k, i \in (a_k) \implies (i) \subset (a_k)$.

The chain in question produces an ascending chain of multisets of irreducible factors of a_k, and since $(i) \subset (a_k)$, each multiset is contained in the multiset of irreducible factors of i.

Thus, the multiset chain must stabilize and the ideal chain stabilizes as well. $\square$

2.9

The *height* of a prime ideal P in a ring R is (if finite) the maximum length h of a chain of prime ideals $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_h = P$ in R. (Thus, the Krull dimension of R, if finite, is the maximum height of a prime ideal in R.) Prove that if R is a UFD, then every prime ideal of height 1 in R is principal.

Solution.
Suppose $P \subset R$ is a prime ideal s.t. all irreducible elements in P are associates. Then $P = (p)$ for any irreducible $p \in P$. Indeed, fix an irreducible $p \in P$ and consider any $a \in P$. Since R is a UFD, a has a unique factorization $a = p_1 \dots p_n \in P \implies p_k \in P$ for some k. Since all irreducibles are associate in P, we have $p \mid p_k \implies p \mid a$, so $a = a'p \in (p)$. Thus, $P \subset (p)$ and also $(p) \subset P \implies P = (p)$.

Now if a prime ideal P contains 2 non-associate irreducibles p and q, then $(0) \subsetneq (p) \subsetneq P$, so the height of P is at least 2.

If P is a non-zero prime ideal, then it contains at least one irreducible element: if $a \in P$, $a \neq 0$, then P must contain an irreducible factor of a. It follows that prime ideals of height 1 are principal. $\square$

2.10

It is a consequence of a theorem known as *Krull's Hauptidealsatz* that every nonzero, nonunit element in a Noetherian domain is contained in a prime ideal of height 1. Assuming this, prove a converse to Exercise 2.9, and conclude that a Noetherian domain R is a UFD if and only if every prime ideal of height 1 in R is principal.

Solution.
We will use the characterization theorem for UFDs: an integral domain R is a UFD iff

- The a.c.c. for principal ideals holds in R
- Irreducibles are prime in R

If R is a Noetherian domain such that every prime ideal of height 1 is principal, then the a.c.c. holds, so we only need to show that irreducibles are primes.

Let $p \in R$ be irreducible. It is contained in a prime ideal $P = (q)$ of height 1. $p = rq \implies r$ is a unit, hence $(p) = (q) = P$ is prime, thus p is prime. $\square$

2.11

Let R be a PID, and let I be a nonzero ideal of R. Show that R/I is an artinian ring (cf. Exercise 1.10), by proving explicitly that the d.c.c. holds in R/I.

Solution.
The descending chain of ideals in R/I naturally corresponds to a descending chain of principal ideals containing I in R. The latter stabilizes, as proved in 2.8. □

2.12

Prove that if $R[x]$ is a PID, then R is a field.

Solution.
Suppose $(x) \subset I$, where I is an ideal of $R[x]$. Since $R[x]$ is a PID, $I = (f(x))$ for some $f(x)$. R is an integral domain, because $R[x]$ is. We have $x = g(x)f(x) \implies \deg g + \deg f = 1 \implies g(x) = ax + b, f(x) = c$ or $g(x) = c, f(x) = ax + b$.

In the first case we have $x = acx + bc \implies c$ is a unit and $(f) = (1)$.

In the second case a is a unit and $b = 0 \implies (f) = (ax) = (x)$.

Hence, (x) is maximal and $R \simeq \frac{R[x]}{(x)}$ is a field.

Alternatively, we can show that (x) is prime as follows: suppose $fg \in (x) \implies f(x)g(x) = h(x)x$. Evaluate the expression at $x = 0$: $f(0)g(0) = 0$. Since R is an integral domain, $f(0) = 0$ or $g(0) = 0$. If $f(0) = 0$, then $f(x) \in (x)$, so (x) is indeed prime. Since primes are maximal in PIDs, $R \simeq \frac{R[x]}{(x)}$ is a field.

Alternatively, $\frac{R[x]}{(x)} \simeq R$ is an integral domain, thus (x) is prime in $R[x]$, and primes are maximal in PIDs, therefore R is a field. □

2.15

Prove that if R is a Euclidean domain, then R admits a Euclidean valuation $\overline{v}$ such that $\overline{v}(ab) \geq \overline{v}(b)$ for all nonzero $a, b \in R$. (Hint: Since R is a Euclidean domain, it admits a valuation v as in Definition 2.7. For $a \neq 0$, let $\overline{v}(a)$ be the minimum of all $v(ab)$ as $b \in R$, $b \neq 0$. To see that R is a Euclidean domain with respect to $\overline{v}$ as well, let a, b be nonzero in R, with $b \nmid a$; choose q, r so that $a = bq + r$, with $v(r)$ minimal; assume that $\overline{v}(r) \geq \overline{v}(b)$, and get a contradiction.)

Solution.
$\overline{v}(b) = \min_{t \in R \setminus \{0\}} v(bt) = v(by)$ for some $y \in R$. Divide a by by with remainder:

$$a = q'by + r', \quad v(r') < v(by)$$

Since $a = (q'y)b + r'$, we must have $v(r') \geq v(r)$. The desired contradiction is:

$$\overline{v}(b) = v(by) > v(r') \geq v(r) \geq \overline{v}(r) \geq \overline{v}(b)$$

Note that the inequality $\overline{v}(ab) \geq \overline{v}(b)$ holds, since $\overline{v}(ab) = v(abx)$ for some x and $v(abx) \geq \overline{v}(b)$. $\square$

2.16

Let R be a Euclidean domain with Euclidean valuation v; assume that $v(ab) \geq v(b)$ for all nonzero $a, b \in R$ (cf. Exercise 2.15). Prove that associate elements have the same valuation and that units have minimum valuation.

Solution.
If a and b are associates, then $a = ub$, $b = va$, where u, v are units. Then

$$v(a) = v(ub) \geq v(b) = v(va) \geq v(a) \implies v(a) = v(b)$$

If u is a unit, then $\forall r \in R$

$$v(r) = v(uu^{-1}r) \geq v(u)$$

$\square$

2.17

Let R be a Euclidean domain that is not a field. Prove that there exists a nonzero, nonunit element c in R such that $\forall a \in R$, $\exists q, r \in R$ with $a = qc + r$ and either $r = 0$ or r a unit.

Solution.
Let $c \in R$ be a nonzero nonunit element with minimal valuation. Then $\forall a \in R \quad a = qc + r$ with $r = 0$ or $v(r) < v(c) \implies r$ is a unit. $\square$

2.18

For an integer d, denote by $\mathbb{Q}(\sqrt{d})$ the smallest subfield of $\mathbb{C}$ containing $\mathbb{Q}$ and $\sqrt{d}$, with norm N defined as in Exercise III.4.10. See Exercise 1.17 for the case $d = -5$; in this problem, you will take $d = -19$.

Let $\delta = (1 + i\sqrt{19})/2$, and consider the following subring of $\mathbb{Q}(\sqrt{-19})$:

$$\mathbb{Z}[\delta] := \left\{ a + b\frac{1 + i\sqrt{19}}{2} \mid a, b \in \mathbb{Z} \right\}.$$

- Prove that the smallest values of $N(z)$ for $z = a + b\delta \in \mathbb{Z}[\delta]$ are $0, 1, 4, 5$. Prove that $N(a + b\delta) \geq 5$ if $b \neq 0$.
- Prove that the units in $\mathbb{Z}[\delta]$ are ± 1.
- If $c \in \mathbb{Z}[\delta]$ satisfies the condition specified in Exercise 2.17, prove that c must divide 2 or 3 in $\mathbb{Z}[\delta]$, and conclude that $c = \pm 2$ or $c = \pm 3$.
- Now show that $\nexists q \in \mathbb{Z}[\delta]$ such that $\delta = qc + r$ with $c = \pm 2, \pm 3$ and $r = 0, \pm 1$.

Conclude that $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is not a Euclidean domain.

Solution.
$N(a+bi\sqrt{19}) = a^2+19b^2, \quad N(a+b\delta) = N(a+\frac{b}{2}+\frac{b}{2}i\sqrt{19}) = a^2+ab+\frac{b^2}{4}+19\frac{b^2}{4} = a^2+ab+5b^2 \in \mathbb{N}.$

If $b \neq 0$, then $N(a + b\delta) = (a + \frac{b}{2})^2 + \frac{19}{4}b^2 \geq \frac{19}{4}$. Furthermore, this implies $N(a + b\delta) \geq 5$, since $N(a + b\delta) \in \mathbb{N}$.

If $b = 0$, then $N(a + b\delta) = a^2$ and the minimal values are $0, 1, 4$. Note that $N(0 + 1\delta) = 5$.

If $u \in \mathbb{Z}[\delta]$ is a unit, then $N(1) = N(uu^{-1}) = N(u)N(u^{-1}) \implies N(u) \mid N(1) = 1 \implies N(u) = 1$. Since $N(a + b\delta) \geq 5$ if $b \neq 0$, $N(u) = 1$ is only possible if $u = a + 0\delta \implies N(u) = a^2 = 1 \implies a = \pm 1$. Thus, the only units in $\mathbb{Z}[\delta]$ are ± 1.

Assume there is nonzero nonunit $c \in \mathbb{Z}[\delta]$ s.t. $\forall x \in \mathbb{Z}[\delta]\ \exists q, r \in \mathbb{Z}[\delta]$ s.t. $x = qc + r$ with $r = 0$ or r a unit. Take $x = 2$: $2 = qc + r$. If $r = 0$, then $c \mid 2$. If $r \neq 0$, then r is a unit: $r = \pm 1$. In the case $r = 1$ we get $2 = qc + 1 \implies 1 = qc \implies c$ is a unit, which is not the case. Hence, we must have $2 = qc - 1 \implies 3 = qc \implies c \mid 3$. In any case, $c \mid 2$ or $c \mid 3$.

If $c \mid 2$, then $N(c) \mid N(2) = 4 \implies N(c) = 4$ and $c = \pm 2$ If $c \mid 3$, then $N(c) \mid 9 \implies N(c) = 9$ and $N(q) = 1 \implies q$ is a unit and $c = \pm 3$.

Suppose now $\exists q \in \mathbb{Z}[\delta]$ s.t. $\delta = qc + r$ with $c = \pm 2, \pm 3$ and $r = 0, \pm 1$.

$$N(\delta) = 5, \quad N(\delta - 1) = N\left(-\frac{1}{2} + \frac{1}{2}i\sqrt{19}\right) = 5, \quad N(\delta + 1) = N\left(\frac{3}{2} + \frac{1}{2}i\sqrt{19}\right) = 7.$$

Now it is clear that such q can not exist: for $r = 0, -1, 1$ we must have $N(c) \mid N(\delta)$, $N(c) \mid N(\delta+1)$ or $N(c) \mid N(\delta - 1)$ respectively. Since $N(c)$ is either 4 or 9, this is not possible. Thus, $\mathbb{Z}[\delta]$ is not a Euclidean domain. □

2.19

A *discrete valuation* on a field k is a surjective homomorphism of abelian groups $v : (k^*, \cdot) \to (\mathbb{Z}, +)$ such that $v(a + b) \geq \min(v(a), v(b))$ for all $a, b \in k^*$ such that $a + b \in k^*$.

- Prove that the set $R := \{a \in k^* \mid v(a) \geq 0\} \cup \{0\}$ is a subring of k.

- Prove that R is a Euclidean domain.

Rings arising in this fashion are called *discrete valuation rings*, abbreviated DVR. They arise naturally in number theory and algebraic geometry. Note that the Krull dimension of a DVR is 1 (Example III.4.14); in algebraic geometry, DVRs correspond to particularly nice points on a 'curve'.

- Prove that the ring of rational numbers a/b with b not divisible by a fixed prime integer p is a DVR.

Solution.
Note that $0 = v(1) = v((-1)^2) = 2v(-1) \implies v(-1) = 0 \implies \forall a \in k^*, v(-a) = v(-1) + v(a) = v(a)$. Now it is clear that R is a subring. If $a, b \in R$, $a, b \neq 0$, $a \neq -b$, then $v(-a) = v(a) \geq 0 \implies -a \in R$. $v(a+b) \geq \min(v(a), v(b)) \geq 0 \implies a+b \in R$. $v(ab) = v(a) + v(b) \geq 0 \implies ab \in R$. If $b = -a$, then $a + b = 0 \in R$. If $b = 0$, then $-b = 0 \in R$, $a + b = a \in R$, $ab = 0 \in R$.

Division with remainder in R is quite trivial. Consider any $a, b \in R$. If $v(a) < v(b)$, then $a = 0b + a$. Otherwise, $a = (ab^{-1})b + 0$, and $v(ab^{-1}) = v(a) - v(b) \geq 0$, so indeed $ab^{-1} \in R$.

Consider the field $\mathbb{Q}$ with $v\left(\frac{a}{b}\right) = \operatorname{ord}_p(a) - \operatorname{ord}_p(b)$.

$$\begin{aligned}
v\left(\frac{a}{b}\frac{a'}{b'}\right) = v\left(\frac{aa'}{bb'}\right) &= \operatorname{ord}_p(aa') - \operatorname{ord}_p(bb') \\
&= \operatorname{ord}_p(a) - \operatorname{ord}_p(b) + \operatorname{ord}_p(a') - \operatorname{ord}_p(b') \\
&= v\left(\frac{a}{b}\right) + v\left(\frac{a'}{b'}\right),
\end{aligned}$$

so $v : \mathbb{Q}^* \to \mathbb{Z}$ is indeed a homomorphism.

$$\begin{aligned}
v\left(\frac{a}{b} + \frac{a'}{b'}\right) = v\left(\frac{ab' + a'b}{bb'}\right) &= \operatorname{ord}_p(ab' + a'b) - \operatorname{ord}_p(bb') \\
&\geq \min(\operatorname{ord}_p(ab'), \operatorname{ord}_p(a'b)) - \operatorname{ord}_p(bb') \\
&= \min(\operatorname{ord}_p(ab') - \operatorname{ord}_p(bb'), \operatorname{ord}_p(a'b) - \operatorname{ord}_p(bb')) \\
&= \min(\operatorname{ord}_p(a) - \operatorname{ord}_p(b), \operatorname{ord}_p(a') - \operatorname{ord}_p(b')) \\
&= \min\left(v\left(\frac{a}{b}\right), v\left(\frac{a'}{b'}\right)\right),
\end{aligned}$$

so the inequality $v(a+b) \geq \min(v(a), v(b))$ is satisfied and v is a discrete valuation.

Now if $\frac{a}{b}$ is in reduced form, then $\operatorname{ord}_p(a) = 0$ or $\operatorname{ord}_p(b) = 0$, thus $v\left(\frac{a}{b}\right) \geq 0 \iff \operatorname{ord}_p(a) - \operatorname{ord}_p(b) \geq 0 \iff \operatorname{ord}_p(b) = 0$, so the DVR R in this case is exactly a ring of numbers $\frac{a}{b}$ with b not divisible by p. $\square$

2.20

As seen in Exercise 2.19, DVRs are Euclidean domains. In particular, they must be PIDs. Check this directly, as follows. Let R be a DVR, and let $t \in R$ be an element such that $v(t) = 1$. Prove that if $I \subseteq R$ is any nonzero ideal, then $I = (t^k)$ for some $k \geq 1$. (The element t is called a 'local parameter' of R.)

Solution.
Let k be a field, v a discrete valuation and R the corresponding DVR. Note that units and only units in R have zero valuation. Indeed, if u is a unit in R, then $u^{-1} \in R \implies v(u^{-1}) = -v(u) \geq 0 \implies 0 \leq v(u) \leq 0 \implies v(u) = 0$. On the other hand, if $v(u) = 0$, then $\exists u^{-1} \in k$ and $v(u^{-1}) = -v(u) = 0 \geq 0 \implies u^{-1} \in R$, so u is a unit in R.

It follows that $a, b \in R$ are associates $\iff v(a) = v(b)$.

Now let I be a proper nonzero ideal of R and let $a \in I$ be an element with minimal valuation. Necessarily, $v(a) \geq 1$, for otherwise I would contain a unit. We have then $\forall i \in I$, $i = (ia^{-1})a$ and $ia^{-1} \in R$, since $v(i) \geq v(a) \implies I = (a)$. Furthermore, if $t \in R$ s.t. $v(t) = 1$, then denote $v(a)$ by k to get $v(a) = k \cdot 1 = kv(t) = v(t^k) \implies (a) = (t^k)$ with $k \geq 1$. □

2.21

Prove that an integral domain is a PID if and only if it admits a Dedekind-Hasse valuation. (Hint: For the $\Longleftarrow$ implication, adapt the argument in Proposition 2.8; for $\Longrightarrow$, let $v(a)$ be the size of the multiset of irreducible factors of a.)

Solution.
$\Longleftarrow$ Let R be an integral domain, v a Dedekind-Hasse valuation and $I \subset R$ an ideal. Let $a \in I$ be an element with minimal valuation. $\forall i \in I$ either $a \mid i$ or $\exists r \in (a, i)$ with $v(r) < v(a)$. The second case is not possible, since $r \in (a, i) \subset I$ contradicts the minimality of $v(a)$. Hence $\forall i \in I$, $a \mid i$, which means $I = (a)$.

$\Longrightarrow$ Let R be a PID and $v(a)$ be the size of the multiset of irreducible factors of a. Consider any $a, b \in R$, $b \neq 0$, $b \nmid a$. Since R is a PID, $(a, b) = (d)$ for some $d \in R$. I claim that $v(d) < v(b)$. Indeed, $(b) \subsetneq (a, b) = (d)$, therefore, the multiset of irreducible factors of d is strictly contained in the multiset of irreducible factors of b, hence $v(d) < v(b)$. □

2.22

Suppose $R \subseteq S$ is an inclusion of integral domains, and assume that R is a PID. Let $a, b \in R$, and let $d \in R$ be a gcd for a and b in R. Prove that d is also a gcd for a and b in S. [5.2]

Solution.
Suppose $d' \in S$, $d' \mid a$, $d' \mid b$. Since R is a PID, $d = ax + by$ for some $x, y \in R \implies d' \mid d$. Thus, d is a gcd of a, b in S. □

VI: Linear Algebra

§1 Free modules revisited

1.7

Let R be an integral domain, and let $M = R^{\oplus A}$ be a free R-module. Let K be the field of fractions of R, and view M as a subset of $V = K^{\oplus A}$ in the evident way. Prove that a subset $S \subseteq M$ is linearly independent in M (over R) if and only if it is linearly independent in V (over K). Conclude that the rank of M (as an R-module) equals the dimension of V (as a K-vector space). Prove that if S generates M over R, then it generates V over K. Is the converse true?

Solution.
Let $R \hookrightarrow K$ via $r \mapsto \frac{r}{1}$. This induces an inclusion $R^{\oplus A} \hookrightarrow K^{\oplus A}$ defined by:

$$\sum r_i a_i \mapsto \sum \frac{r_i}{1} a_i$$

$\Longrightarrow$ Assume $S \subset M$ is linearly independent over R. Suppose we have a linear relation in V:

$$\sum_{i \in I} \frac{r_i}{p_i} s_i = 0$$

where $|I| < +\infty$. Then:

$$\sum_{i \in I} P \frac{r_i}{p_i} s_i = 0, \quad \text{where } P = \prod_{i \in I} p_i$$

For all $k \in I$, the term $\frac{P}{p_k} = \prod_{i \in I \setminus \{k\}} p_i$ is an element of R. Hence, we have obtained a linear relation in M. Since S is linearly independent in M, all coefficients must be zero, which implies $r_i = 0$ for all $i \in I$.

$\Longleftarrow$ Obvious.

Suppose that S generates M over R and let $v \in V$. Then we can write v as a finite sum:

$$v = \sum_{i \in I} \frac{r_i}{p_i} a_i$$

Let $P = \prod_{i \in I} p_i$. Multiplying by P clears the denominators, meaning $Pv \in M$. Since S generates M, there exist $r_j \in R$ and $s_j \in S$ such that:

$$Pv = \sum_{j \in J} r_j s_j$$

Dividing by P in V, we get:

$$v = \sum_{j \in J} \frac{r_j}{P} s_j \in \langle S \rangle_K$$

Thus, S generates V over K. The converse is not true. For example, the subset $\{2\}$ generates $\mathbb{Q}$ over $\mathbb{Q}$, but does not generate $\mathbb{Z}$ over $\mathbb{Z}$. $\square$

1.9

Let R be a commutative ring, and let M be an R-module. Let $\mathfrak{m}$ be a maximal ideal in R, such that $\mathfrak{m}M = 0$ (that is, $rm = 0$ for all $r \in \mathfrak{m}$, $m \in M$). Define in a natural way a vector space structure over $R/\mathfrak{m}$ on M.

Solution.
We define scalar multiplication by $\overline{r}m := rm$. We must check that this operation is correctly defined. If $\overline{r} = \overline{r}'$, then:

$$r - r' \in \mathfrak{m} \implies (r - r')m = 0 \implies rm = r'm$$

Thus, the operation is well-defined. The other vector space axioms trivially hold because M is an R-module. $\square$

1.10

Let R be a commutative ring, and let F be a free R-module with basis B. Let $\mathfrak{m}$ be a maximal ideal of R, and let $k = R/\mathfrak{m}$. Prove that $F/\mathfrak{m}F \simeq k^{\oplus B}$ as k-vector spaces.

Solution.
Let $\sum r_i b_i \in F$. If $x \in \mathfrak{m}F$, then $x = \sum m_i f_i = \sum m_i \sum r_{ij} b_j = \sum m_i' b_i$, where $m_i' \in \mathfrak{m}$. On the other hand, all $\sum m_i b_i$ with $m_i \in \mathfrak{m}$ are in $\mathfrak{m}F$. Hence, $\mathfrak{m}F = \mathfrak{m}^{\oplus B}$.

$F/\mathfrak{m}F$ is an R-module, and for all $m \in \mathfrak{m}$ and for all $\overline{f} \in F/\mathfrak{m}F$:

$$m\overline{f} = \overline{mf} = \overline{0}$$

Hence, by Exercise 1.9, $F/\mathfrak{m}F$ is a k-vector space.

Now let's prove that $\pi(B)$ is a basis of $F/\mathfrak{m}F$, where π is the natural projection. Obviously, $\pi(B)$ generates $F/\mathfrak{m}F$ since B generates F. Suppose that $\sum_I (r_i + \mathfrak{m})(b_i + \mathfrak{m}F) = 0 + \mathfrak{m}F$. Then:

$$\left(\sum r_i b_i\right) + \mathfrak{m}F = \overline{0} \implies \sum r_i b_i \in \mathfrak{m}F \implies r_i \in \mathfrak{m} \implies r_i + \mathfrak{m} = \overline{0}$$

This implies that $\pi(B)$ is linearly independent, hence it is a basis. Therefore, $F/\mathfrak{m}F \simeq k^{\oplus B}$. $\square$

1.11

Prove that commutative rings satisfy the IBN (Invariant Basis Number) property.

Solution.
Let R be a commutative ring which has a proper ideal and let F be a free R-module. Suppose $F \simeq R^{\oplus A} \simeq R^{\oplus B}$.

R has a maximal ideal $\mathfrak{m}$ and by Exercise 1.10,

$$k^{\oplus A} \simeq F/\mathfrak{m}F \simeq k^{\oplus B}$$

where $k = R/\mathfrak{m}$. But then it follows that $A \simeq B$, since k is a field. $\square$

1.13

Let A be an abelian group such that $\mathrm{End}_{\mathrm{Ab}}(A)$ is a field of characteristic 0. Prove that $A \simeq \mathbb{Q}$.

Solution.
$\mathrm{End}(A)$ acts on A: $\varphi a := \varphi(a)$ and makes the latter into a vector space.

We have an inclusion $\mathbb{Q} \hookrightarrow \mathrm{End}(A)$, since $\mathrm{char}(\mathrm{End}\, A) = 0$:

$$\frac{p}{q} \mapsto (p\,\mathrm{id}) \circ (q\,\mathrm{id})^{-1}$$

Hence, A is a $\mathbb{Q}$-vector space via

$$\frac{p}{q}a = ((p\,\mathrm{id}) \circ (q\,\mathrm{id})^{-1})(a) = (q\,\mathrm{id})^{-1}(pa)$$

Basically, $p \cdot a = (p\,\mathrm{id})(a) = pa$ for integer p, and $\frac{1}{q}a = b$ such that $a = qb$. It exists and is unique because $q\,\mathrm{id}$ has an inverse, hence is an isomorphism.

Now, suppose $f \in \mathrm{End}_{\mathrm{Ab}}(A)$. Then

$$\forall n \in \mathbb{Z} \quad f(na) = nf(a)$$

We have also $f(a) = mf\left(\frac{1}{m}a\right) \implies f\left(\frac{1}{m}a\right) = \frac{1}{m}f(a)$ for $m \in \mathbb{N}$.

Hence, $f\left(\frac{n}{m}a\right) = nf\left(\frac{1}{m}a\right) = \frac{n}{m}f(a)$ or simply

$$f(ra) = rf(a) \quad \forall r \in \mathbb{Q}$$

This means that f is actually a $\mathbb{Q}$-vector space endomorphism, so $\mathrm{End}_{\mathrm{Ab}}(A) = \mathrm{End}_{\mathbb{Q}\text{-Vect}}(A)$.

Suppose S is a basis of A, $|S| > 1$ and $a \in S$. Then $A = V \oplus W$, where $V = \langle a \rangle$ and $W = \langle S \backslash \{a\} \rangle$. Define a map $\sigma : A \to A$ via

$$A \simeq V \oplus W \xrightarrow{\pi_1} V \hookrightarrow A$$

which sends $\lambda a + \sum_{s \in S \setminus \{a\}} \lambda_s s$ to λa.

It is a $\mathbb{Q}$-vector space homomorphism as a composition of homomorphisms and has a non-trivial kernel, since $S \setminus \{a\} \subset \ker \sigma$ and $S \setminus \{a\}$ is non-empty and does not contain 0. Hence, σ does not have an inverse, contradicting the fact that $\mathrm{End}(A)$ is a field.

It follows that $|S| = \dim A \leq 1$. The case of 0 is not possible, so $\dim_{\mathbb{Q}} A = 1$ and $A \simeq \mathbb{Q}$.

We can also show that for a $\mathbb{Q}$-vector space V, $\mathrm{End}_{\mathbb{Q}\text{-Vect}}(V) \simeq M_n(\mathbb{Q})$ as rings, where $n = \dim V$. Thus, in the general case, $\mathrm{End}_{\mathrm{Ab}}(V)$ is not a field, and $\mathbb{Q}$ is the only space with its endomorphism ring being a field. □

1.19

Let k be a field, and let $f(x) \in k[x]$ be any polynomial. Prove that there exists a multiple of $f(x)$ in which all exponents of nonzero monomials are *prime* integers. (Example: for $f(x) = 1+x^5+x^6$, $(1 + x^5 + x^6)(2x^2 - x^3 + x^5 - x^8 + x^9 - x^{10} + x^{11}) = 2x^2 - x^3 + x^5 + 2x^7 + 2x^{11} - x^{13} + x^{17}$.)

Solution.
Let $V = k[x]/(f(x))$. This is a k-vector space with basis $\overline{1}, \overline{x}, \ldots, \overline{x^{n-1}}$, where $n = \deg f$.

Take $n + 1$ distinct primes $p_1, \ldots, p_{n+1}$. The system

$$S = \{\overline{x^{p_1}}, \overline{x^{p_2}}, \ldots, \overline{x^{p_{n+1}}}\}$$

is linearly dependent, since $|S| = n + 1 > \dim V$. So, for some $\lambda_i \in k$ (not all zero):

$$\sum_{i=1}^{n+1} \lambda_i \overline{x^{p_i}} = \overline{0} \implies \sum_{i=1}^{n+1} \lambda_i x^{p_i} \in (f(x))$$

$$\implies \exists p(x) \in k[x] \quad \text{such that} \quad f(x)p(x) = \sum_{i=1}^{n+1} \lambda_i x^{p_i}$$

This multiple has only prime exponents, as desired. □

§2 Homomorphisms of free modules, I

§3 Homomorphisms of free modules, II

3.6

Let R be a commutative ring and $M = \langle m_1, \ldots, m_r \rangle$ a finitely generated R-module. Let $A \in M_r(R)$ be a matrix such that

$$A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Prove that $\det(A)m = 0$ for all $m \in M$.

Solution.
Let A^* be the adjoint matrix of A. Recall the fundamental property $A^*A = \det(A)I_r$. Starting from the given equation:

$$A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Multiply both sides on the left by A^*:

$$A^*A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = A^* \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\det(A)I_r \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

This implies $\det(A)m_i = 0$ for every generator m_i. Since any $m \in M$ is an R-linear combination of these generators, $\det(A)m = 0$ for all $m \in M$. □

3.7

Let R be a commutative ring, M a finitely generated R-module, and let J be an ideal of R. Assume $JM = M$. Prove that there exists an element $b \in J$ such that $(1 + b)M = 0$.

Solution.
Since $M = JM$, each generator m_i can be written as $m_i = \sum_{j=1}^{r} b_{ij}m_j$ with $b_{ij} \in J$. In matrix form, this is $\mathbf{m}^T = B\mathbf{m}^T$, or $(I - B)\mathbf{m}^T = 0$. By Exercise 3.6, $\det(I - B)$ annihilates M.

Expanding the determinant $\det(I - B)$ modulo J, we see that since all entries of B are in J:

$$\det(I - B) \equiv \det(I) \equiv 1 \pmod{J}$$

Thus, $\det(I - B) = 1 + b$ for some $b \in J$. Hence $(1 + b)M = 0$. □

3.8

Let R be a commutative ring, M be a finitely generated R-module, and let J be an ideal of R contained in the Jacobson radical of R (Exercise V.3.14). Prove that $M = 0 \iff JM = M$.

Solution.
$\Longrightarrow$ If $M = 0$, then $JM = 0 = M$.

$\Longleftarrow$ Assume $JM = M$. By Exercise 3.7, there exists $b \in J$ such that $(1 + b)M = 0$. Since J is contained in the Jacobson radical, $1 + b$ is a unit in R by Exercise V.3.14. Hence, $M = (1 + b)^{-1}(1 + b)M = 0$. □

3.9

Let $(R, \mathfrak{m})$ be a commutative local ring (meaning a ring with a single maximal ideal $\mathfrak{m}$) and M, N be finitely generated R-modules. Prove that if $M = \mathfrak{m}M + N$, then $M = N$.

Solution.
Since M is a finitely generated R-module, its quotient is also finitely generated. For any ideal $I \subseteq R$ and submodule $N \subseteq M$, we have

$$I(M/N) = \frac{IM + N}{N}$$

Applying this to the maximal ideal $\mathfrak{m}$, we have:

$$\mathfrak{m}(M/N) = \frac{\mathfrak{m}M + N}{N} = M/N$$

In a local ring, the Jacobson radical $J(R)$ is equal to the unique maximal ideal $\mathfrak{m}$. Therefore, we have $J(R)(M/N) = M/N$. Since M/N is finitely generated, Nakayama's Lemma implies that $M/N = 0$. Hence, $M = N$. □

3.10

Let R be a commutative local ring, and let M be a finitely generated R-module. Note that $M/\mathfrak{m}M$ is a finite-dimensional vector space over the field $R/\mathfrak{m}$; let $m_1, \dots, m_r \in M$ be elements whose cosets mod $\mathfrak{m}M$ form a basis of $M/\mathfrak{m}M$. Prove that $m_1, \dots, m_r$ generate M.

Solution.
$M/\mathfrak{m}M$ is a vector space over $R/\mathfrak{m}$ by 1.9, and $\dim M/\mathfrak{m}M < +\infty$ since M is finitely generated as an R-module.

Suppose $m \in M$. Since $m_1, \ldots, m_r$ is a basis of $M/\mathfrak{m}M$, we have

$$m + \mathfrak{m}M = \sum (s_i + \mathfrak{m})(m_i + \mathfrak{m}M) = \left(\sum s_i m_i\right) + \mathfrak{m}M$$
$$\implies m = \sum s_i m_i + m', \quad \text{where } m' \in \mathfrak{m}M$$

Thus, $M = \langle m_1, \ldots, m_r \rangle + \mathfrak{m}M$ and by 3.9

$$M = \langle m_1, \ldots, m_r \rangle$$

□

3.16

Every complex $V_\bullet$ determines an element in $K(k\text{-Vect}^f)$ (Grothendieck group), namely

$$\chi_K(V_\bullet) := \sum_i (-1)^i [V_i] \in K(k\text{-Vect}^f).$$

- χ_K 'is an Euler characteristic', in the sense that it satisfies the formula given in Proposition 3.13: $\chi_K(V_\bullet) = \sum_i (-1)^i [H_i(V_\bullet)]$.
- χ_K is a 'universal Euler characteristic', in the following sense. Let G be an abelian group, and let δ be a function associating an element of G to each finite-dimensional vector space, such that $\delta(V) = \delta(V')$ if $V \simeq V'$ and $\delta(V/U) = \delta(V) - \delta(U)$. For $V_\bullet$ a complex, define $\chi_G(V_\bullet) = \sum_i (-1)^i \delta(V_i)$. Then δ induces a (unique) group homomorphism $K(k\text{-Vect}^f) \to G$ mapping $\chi_K(V_\bullet)$ to $\chi_G(V_\bullet)$.
- In particular, $\delta = \dim$ induces a group homomorphism $K(k\text{-Vect}^f) \to \mathbb{Z}$ such that $\chi_K(V_\bullet) \mapsto \chi(V_\bullet)$.
- This is in fact an isomorphism.

Solution.
Part 1: χ_K is an Euler characteristic. We prove the formula $\chi_K(V_\bullet) = \sum_i (-1)^i [H_i(V_\bullet)]$ by induction on the length n of the complex $V_\bullet$.

Base case $n = 0$: Consider the complex $0 \to V_0 \to 0$. We have $H_0(V_\bullet) = V_0$. Thus:

$$\chi_K(V_\bullet) = [V_0] = [H_0(V_\bullet)]$$

Base case $n = 1$: Consider $0 \to V_1 \xrightarrow{\alpha} V_0 \to 0$. The homologies are $H_1(V_\bullet) = \ker \alpha$ and $H_0(V_\bullet) = V_0 / \operatorname{im} \alpha$. We have the short exact sequences:

$$0 \to \ker \alpha \to V_1 \to \operatorname{im} \alpha \to 0 \implies [V_1] = [\ker \alpha] + [\operatorname{im} \alpha]$$

$$0 \to \operatorname{im}\alpha \to V_0 \to V_0/\operatorname{im}\alpha \to 0 \implies [V_0] = [\operatorname{im}\alpha] + [V_0/\operatorname{im}\alpha]$$

Computing the Euler characteristic:

$$\chi_K(V_\bullet) = [V_0] - [V_1] = ([\operatorname{im}\alpha] + [V_0/\operatorname{im}\alpha]) - ([\ker\alpha] + [\operatorname{im}\alpha]) = [V_0/\operatorname{im}\alpha] - [\ker\alpha]$$

$$= [H_0(V_\bullet)] - [H_1(V_\bullet)] = \sum_{i=0}^{1}(-1)^i[H_i(V_\bullet)]$$

Inductive step: Assume the statement holds for complexes of length n. Consider a complex $V_\bullet$ of length $n+1$:

$$0 \to V_{n+1} \xrightarrow{\alpha_{n+1}} V_n \xrightarrow{\alpha_n} \dots \xrightarrow{\alpha_1} V_0 \to 0$$

Let $V'_\bullet$ be its truncated complex $0 \to V_n \xrightarrow{\alpha_n} \cdots \to V_0 \to 0$. By the induction hypothesis, $\chi_K(V'_\bullet) = \sum_{i=0}^{n}(-1)^i[H_i(V'_\bullet)]$. By definition, $\chi_K(V_\bullet) = \chi_K(V'_\bullet) + (-1)^{n+1}[V_{n+1}]$. Let us compare the homologies of $V_\bullet$ and $V'_\bullet$:

- $H_i(V_\bullet) = H_i(V'_\bullet)$ for all $i < n$.
- $H_n(V'_\bullet) = \ker\alpha_n$.
- $H_n(V_\bullet) = \ker\alpha_n/\operatorname{im}\alpha_{n+1}$.
- $H_{n+1}(V_\bullet) = \ker\alpha_{n+1}$.

The exact sequence $0 \to \ker\alpha_{n+1} \to V_{n+1} \to \operatorname{im}\alpha_{n+1} \to 0$ gives $[V_{n+1}] = [\ker\alpha_{n+1}] + [\operatorname{im}\alpha_{n+1}]$. The exact sequence $0 \to \operatorname{im}\alpha_{n+1} \to \ker\alpha_n \to \ker\alpha_n/\operatorname{im}\alpha_{n+1} \to 0$ gives $[\ker\alpha_n] = [\operatorname{im}\alpha_{n+1}] + [\ker\alpha_n/\operatorname{im}\alpha_{n+1}]$, which rewrites as $[H_n(V'_\bullet)] = [\operatorname{im}\alpha_{n+1}] + [H_n(V_\bullet)]$.

Now, sum the alternating homologies for $V_\bullet$:

$$\sum_{i=0}^{n+1}(-1)^i[H_i(V_\bullet)] = \sum_{i=0}^{n-1}(-1)^i[H_i(V'_\bullet)] + (-1)^n[H_n(V_\bullet)] + (-1)^{n+1}[H_{n+1}(V_\bullet)]$$

Substitute $[H_n(V_\bullet)] = [H_n(V'_\bullet)] - [\operatorname{im}\alpha_{n+1}]$ and $[H_{n+1}(V_\bullet)] = [\ker\alpha_{n+1}]$:

$$= \sum_{i=0}^{n-1}(-1)^i[H_i(V'_\bullet)] + (-1)^n([H_n(V'_\bullet)] - [\operatorname{im}\alpha_{n+1}]) + (-1)^{n+1}[\ker\alpha_{n+1}]$$

$$= \sum_{i=0}^{n}(-1)^i[H_i(V'_\bullet)] + (-1)^{n+1}([\operatorname{im}\alpha_{n+1}] + [\ker\alpha_{n+1}])$$

$$= \chi_K(V'_\bullet) + (-1)^{n+1}[V_{n+1}] = \chi_K(V_\bullet)$$

This completes the induction.

Part 2: χ_K is a universal Euler characteristic. Recall that $K(k\text{-Vect}^f) = F/E$, where F is the free abelian group on isomorphism classes of finite-dimensional vector spaces, and E is the subgroup generated by relations from short exact sequences. Define a set-map from the generators of F to G via $[V] \mapsto \delta(V)$. Because $\delta(V) = \delta(V')$ whenever $V \simeq V'$, this map is correctly defined and extends linearly to a unique group homomorphism $\varphi : F \to G$.

If $0 \to U \to V \to W \to 0$ is exact, then $W \simeq V/U$. Thus:

$$\varphi([V] - [U] - [W]) = \delta(V) - \delta(U) - \delta(V/U) = 0$$

Hence, the subgroup $E \subset \ker \varphi$. By the universal property of quotients, φ factors through to a unique induced morphism $\varphi^* : K(k\text{-Vect}^f) \to G$ such that $\varphi^*([V]) = \delta(V)$. Applying this to a complex $V_\bullet$:

$$\varphi^*(\chi_K(V_\bullet)) = \varphi^*\left(\sum_i (-1)^i [V_i]\right) = \sum_i (-1)^i \varphi^*([V_i]) = \sum_i (-1)^i \delta(V_i) = \chi_G(V_\bullet)$$

To see φ^* is unique, suppose $\sigma : K(k\text{-Vect}^f) \to G$ also maps $\chi_K(V_\bullet)$ to $\chi_G(V_\bullet)$ for all complexes. Consider the trivial complex $0 \to V \to 0$. Here $\chi_K(V_\bullet) = [V]$ and $\chi_G(V_\bullet) = \delta(V)$. This forces $\sigma([V]) = \delta(V)$, so $\sigma = \varphi^*$.

Part 3 & 4: $\delta = \dim$ induces an isomorphism to $\mathbb{Z}$. Since $\dim(V/U) = \dim V - \dim U$ and dimension is an isomorphism invariant, we apply Part 2. The function $\delta = \dim$ induces a group homomorphism $\varphi : K(k\text{-Vect}^f) \to \mathbb{Z}$ mapping $\chi_K(V_\bullet) \mapsto \chi(V_\bullet)$. We show φ is an isomorphism.

First, observe a key property in $K(k\text{-Vect}^f)$. For any spaces V, U, there is an exact sequence

$$0 \to U \to V \oplus U \to V \to 0$$

It follows that $[V \oplus U] = [V] + [U]$. For any n-dimensional vector space V, $V \simeq k^{\oplus n}$. Therefore, $[V] = n[k] = (\dim V)[k]$.

Surjectivity: The morphism φ maps $[V] \mapsto \dim V$. It is clearly surjective onto $\mathbb{Z}$, as choosing $V = k^{\oplus n}$ yields n.

Injectivity: Let $x \in K(k\text{-Vect}^f)$ be any element. We can write x as a finite linear combination $\sum \lambda_i [V_i]$ for some $\lambda_i \in \mathbb{Z}$. Suppose $x \in \ker \varphi$. Then:

$$0 = \varphi\left(\sum \lambda_i [V_i]\right) = \sum \lambda_i \varphi([V_i]) = \sum \lambda_i \dim V_i$$

Using the identity $[V_i] = (\dim V_i)[k]$ derived above, we rewrite our element x in K:

$$\sum \lambda_i [V_i] = \sum \lambda_i (\dim V_i \cdot [k]) = \left(\sum \lambda_i \dim V_i\right)[k] = 0 \cdot [k] = 0$$

Thus, $\ker \varphi = \{0\}$, meaning φ is injective. Therefore, φ is an isomorphism. $\square$

§4 Presentations and resolutions

4.3

Prove that an integral domain R is a PID if and only if every submodule of R itself is free.

Solution.
$\Longrightarrow$ Let $I \subset R$ be a submodule. By definition, submodules of R are exactly the ideals of R. Since R is a PID, $I = (a)$ for some $a \in R$. If $I = 0$, it is free (with an empty basis). If $I \neq 0$, the map $R \to I$ given by $r \mapsto ra$ is an isomorphism of R-modules. Alternatively, the set $\{a\}$ is linearly independent since R is an integral domain ($ra = 0 \implies r = 0$ for $a \neq 0$), and it generates I. Thus, it is a basis of I, which implies $I \simeq R^1$.

$\Longleftarrow$ Suppose $I \subset R$ is an ideal. By assumption, it is free as an R-module. Every basis of I would be a linearly independent system in R. However, any two non-zero elements $x, y \in R$ are linearly dependent over R, since $y \cdot x - x \cdot y = 0$. Therefore, the maximum size of any linearly independent set in R is 1.

This means $\operatorname{rk} I \leq 1$. Hence, the basis is either empty ($I = 0$) or consists of a single element ($I = (a)$ for some $a \in R$). Since every ideal is generated by at most one element, R is a PID. $\square$

4.4

Let R be a commutative ring and M an R-module.

- Prove that $\operatorname{Ann}(M)$ is an ideal of R.
- If R is an integral domain and M is finitely generated, prove that M is torsion if and only if $\operatorname{Ann}(M) \neq 0$.
- Give an example of a torsion module M over an integral domain, such that $\operatorname{Ann}(M) = 0$. (Of course this example cannot be finitely generated!)

Solution.

$$\operatorname{Ann}(M) = \{r \in R \mid rm = 0 \quad \forall m \in M\}$$

$a \in R, r \in \operatorname{Ann}(M) \implies \forall m \in M, \quad (ar)m = a(rm) = a0 = 0.$

$\forall r, r' \in \operatorname{Ann}(M), \forall m \in M, \quad (r + r')m = rm + r'm = 0 + 0 = 0.$

More is true: $\forall m \in M$ $\operatorname{Ann}(m)$ is an ideal of R and $\operatorname{Ann}(M) = \bigcap_{m \in M} \operatorname{Ann}(m)$

Note that if R is an integral domain and M is finitely generated, there is a surjective morphism $R^{\oplus A} \to M$, $A = \{a_1, \dots, a_n\}$, hence $M \simeq R^{\oplus A}/N$ for some submodule N.

$\Longleftarrow$ If $\operatorname{Ann}(M) \neq 0$, then $\exists r \in R \mid r \neq 0, \quad rm = 0 \quad \forall m \in M \implies \forall m \in M, \quad m$ is torsion.

$\Longrightarrow$ If M is torsion, then $\exists r_i \in R,\ i = 1, \dots, n \mid r_i \overline{a_i} = \overline{0}. \implies \forall m \in M, \quad m = \sum p_i \overline{a_i} \implies rm = 0$, where $r = \prod r_i \neq 0 \implies \operatorname{Ann}(M) \neq 0$.

$\mathbb{Q}$ is not finitely generated as a $\mathbb{Z}$-module. $\mathbb{Z} \subset \mathbb{Q}$ is a submodule. $\mathbb{Q}/\mathbb{Z}$ as a $\mathbb{Z}$-module is torsion:

$$q\left(\frac{p}{q} + \mathbb{Z}\right) = p + \mathbb{Z} = \mathbb{Z}$$

If $n \in \mathbb{Z}$ such that $\forall \overline{s} \in \mathbb{Q}/\mathbb{Z}, \quad n\overline{s} = \overline{0}$, then $\forall m \in \mathbb{N}, \quad n\left(\frac{1}{m} + \mathbb{Z}\right) = \frac{n}{m} + \mathbb{Z} = \mathbb{Z} \implies m \mid n \quad \forall m \in \mathbb{N} \implies n = 0$.

Hence, $\operatorname{Ann}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}) = 0$. □

4.5

Let M be a module over a commutative ring R. Prove that an ideal I of R is the annihilator of an element of M if and only if M contains an isomorphic copy of R/I (viewed as an R-module).

The *associated primes* of M are the prime ideals among the ideals $\operatorname{Ann}(m)$, for $m \in M$. The set of the associated primes of a module M is denoted $\operatorname{Ass}_R(M)$. Note that every prime in $\operatorname{Ass}_R(M)$ contains $\operatorname{Ann}_R(M)$.

Solution.
$\Longrightarrow$ If $I = \operatorname{Ann}(m)$ is an ideal, then $\ker\varphi = I$, where $\varphi : R \to M$ defined by $\varphi(r) = rm$ is an R-module homomorphism. Hence, we have an inclusion $R/I \simeq \operatorname{im}\varphi \hookrightarrow M$.

$\Longleftarrow$ If there is an inclusion $\varphi : R/I \hookrightarrow M$, then consider $m = \varphi(\overline{1})$. If $rm = 0$, then

$$0 = r\varphi(\overline{1}) = \varphi(r\overline{1}) = \varphi(\overline{r}) \implies \overline{r} = \overline{0} \quad (\text{since } \ker\varphi = 0) \implies r \in I$$

This argument works in the reverse direction as well, so $I = \operatorname{Ann}(m)$.

Now, suppose that $\operatorname{Ann}(m)$ is an associated prime. We need to show that $\operatorname{Ann}(m) \supset \operatorname{Ann}(M)$. It is obvious, since

$$\operatorname{Ann}(M) = \bigcap_{m \in M} \operatorname{Ann}(m)$$

□

4.6

Let M be a module over a commutative ring R, and consider the family of ideals $\operatorname{Ann}(m)$, as m ranges over the nonzero elements of M. Prove that the maximal elements in this family are prime ideals of R. Conclude that if R is Noetherian, then $\operatorname{Ass}_R(M) \neq \emptyset$

Solution.
Note that $\forall r \in R, \quad \operatorname{Ann}(m) \subset \operatorname{Ann}(rm)$, for if $sm = 0$, then $s(rm) = r(sm) = 0$.

Suppose $\operatorname{Ann}(m)$ is maximal among annihilators and that $rs \in \operatorname{Ann}(m)$. If $r \notin \operatorname{Ann}(m)$, then $rm \neq 0$.

$s(rm) = 0 \implies s \in \operatorname{Ann}(rm)$. Since $\operatorname{Ann}(m)$ is maximal and $\operatorname{Ann}(m) \subset \operatorname{Ann}(rm)$ (with $rm \neq 0$), we must have $\operatorname{Ann}(m) = \operatorname{Ann}(rm)$. Hence, $s \in \operatorname{Ann}(m)$ and $\operatorname{Ann}(m)$ is prime.

The family of ideals $\mathcal{F} = \{\operatorname{Ann}(m) \mid m \in M \setminus \{0\}\}$ is non-empty.

In a Noetherian ring, every non-empty family of ideals has a maximal element with respect to inclusion. Let P be a maximal element of $\mathcal{F}$. By the argument above, P is a prime ideal. Because $P \in \mathcal{F}$, $P = \operatorname{Ann}(m)$ for some non-zero $m \in M$, meaning P is an associated prime of M.

Therefore, $\operatorname{Ass}_R(M) \neq \emptyset$. □

4.13

Let R be a commutative ring. A tuple $(a_1, a_2, \ldots, a_n)$ of elements of R is a *regular sequence* if a_1 is a non-zero-divisor in R, a_2 is a non-zero-divisor modulo (a_1), a_3 is a non-zero-divisor modulo (a_1, a_2), and so on.

For a, b in R, consider the following complex of R-modules:

$$(*) \qquad 0 \to R \xrightarrow{d_2} R \oplus R \xrightarrow{d_1} R \xrightarrow{\pi} \frac{R}{(a,b)} \to 0$$

where π is the canonical projection, $d_1(r, s) = ra + sb$, and $d_2(t) = (bt, -at)$. Put otherwise, d_1 and d_2 correspond, respectively, to the matrices

$$\begin{pmatrix} a & b \end{pmatrix}, \quad \begin{pmatrix} b \\ -a \end{pmatrix}.$$

- Prove that this is indeed a complex, for every a and b.
- Prove that if (a, b) is a regular sequence, this complex is *exact*.

The complex $(*)$ is called the *Koszul complex* of (a, b). Thus, when (a, b) is a regular sequence, the Koszul complex provides us with a free resolution of the module $R/(a, b)$.

Solution.

$$[d_1 \circ d_2] = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} b \\ -a \end{pmatrix} = 0, \quad \text{so } d_1 \circ d_2 = 0$$

$\pi \circ d_1 = 0$ is obvious, and canonical projection is an epimorphism.

If (a, b) is regular and $d_2(t) = 0$, then

$$(bt, -at) = (0, 0) \implies at = 0 \implies t = 0$$

since a is a non-zero-divisor. Hence, $\ker d_2 = 0$.

If $d_1(r, s) = ra + sb = 0$, then consider this equation modulo (a):

$$\overline{s}\overline{b} = \overline{0} \implies \overline{s} = \overline{0}$$

for $\overline{b}$ is a non-zero-divisor.

$$\implies s = at \quad \text{for some } t \in R.$$

Then

$$0 = ra + atb = a(r + tb) \implies r = -tb$$

Thus, $(r, s) = (-bt, at) = d_2(-t) \in \operatorname{im} d_2 \implies \ker d_1 = \operatorname{im} d_2$.

Finally, $\ker \pi = (a, b) = \operatorname{im} d_1$, even without regularness, and π is an epimorphism. $\square$

4.16

Let $\varphi : R^n \to R^m$ and $\psi : R^p \to R^q$ be two R-module homomorphisms, and let

$$\varphi \oplus \psi : R^n \oplus R^p \to R^m \oplus R^q$$

be the morphism induced on direct sums. Prove that

$$\operatorname{coker}(\varphi \oplus \psi) = \operatorname{coker} \varphi \oplus \operatorname{coker} \psi.$$

Solution.

$$\begin{tikzcd}
R^n \arrow[r, "\varphi"] \arrow[d, "i_1"'] & R^m \arrow[r, "\varphi'"] \arrow[d, "i_1'"'] & \operatorname{coker}\varphi \arrow[d, "i_1''"] \arrow[dr, bend left, "\sigma_1"] & \\
R^n \oplus R^p \arrow[r, "\varphi\oplus\psi"] & R^m \oplus R^q \arrow[r, "\tau"] \arrow[rr, bend right, "\tau^*"'] & \operatorname{coker}\varphi \oplus \operatorname{coker}\psi \arrow[r, dashed, "\lambda"] & M \\
R^p \arrow[u, "i_2"] \arrow[r, "\psi"'] & R^q \arrow[u, "i_2'"'] \arrow[r, "\psi'"'] & \operatorname{coker}\psi \arrow[u, "i_2''"'] \arrow[ur, bend right, "\sigma_2"'] &
\end{tikzcd}$$

Let $v \in R^n$, $w \in R^p$.

$$\tau \circ (\varphi \oplus \psi) \circ i_1(v) = i_1'' \circ \varphi' \circ \varphi(v) = i_1''(0) = (0,0)$$

Similarly, $\tau \circ (\varphi \oplus \psi) \circ i_2 = 0$.

Hence, $\forall (v,w) \in R^n \oplus R^p$,

$$\begin{aligned}\tau \circ (\varphi \oplus \psi)(v,w) &= \tau \circ (\varphi \oplus \psi)(v,0) + \tau \circ (\varphi \oplus \psi)(0,w) \\ &= \tau \circ (\varphi \oplus \psi) \circ i_1(v) + \tau \circ (\varphi \oplus \psi) \circ i_2(w) = (0,0),\end{aligned}$$

so τ indeed annihilates $\varphi \oplus \psi$.

Moreover, if $\tau^* : R^m \oplus R^q \to M$ such that $\tau^* \circ (\varphi \oplus \psi) = 0$, then

$$\tau^* \circ i_2' \circ \psi = \tau^* \circ (\varphi \oplus \psi) \circ i_2 = 0$$

Since $\tau^* \circ i_2'$ annihilates ψ, it induces a unique $\sigma_2 : \operatorname{coker} \psi \to M$ such that

$$\tau^* \circ i_2' = \sigma_2 \circ \psi'$$

Similarly, we obtain $\sigma_1 : \operatorname{coker} \varphi \to M$ such that $\tau^* \circ i_1' = \sigma_1 \circ \varphi'$.

Together σ_1 and σ_2 induce $\lambda : \operatorname{coker} \varphi \oplus \operatorname{coker} \psi \to M$ such that

$$\sigma_1 = \lambda \circ i_1'', \quad \sigma_2 = \lambda \circ i_2''$$

We have then

$$\lambda \circ \tau \circ i_2' = \lambda \circ i_2'' \circ \psi' = \sigma_2 \circ \psi' = \tau^* \circ i_2'$$

and similarly $\lambda \circ \tau \circ i_1' = \tau^* \circ i_1'$.

It follows that $\forall (v,w) \in R^m \oplus R^q$,

$$\begin{aligned}\lambda \circ \tau(v,w) &= \lambda \circ \tau \circ i_1'(v) + \lambda \circ \tau \circ i_2'(w) \\ &= \tau^* \circ i_1'(v) + \tau^* \circ i_2'(w) \\ &= \tau^*(v,w) \implies \tau^* = \lambda \circ \tau\end{aligned}$$

If λ^* is another morphism such that $\tau^* = \lambda^* \circ \tau$, then $\sigma_2 \circ \psi' = \tau^* \circ i_2' = \lambda^* \circ \tau \circ i_2' = \lambda^* \circ i_2'' \circ \psi'$.

Since ψ' is an epimorphism, this would imply $\sigma_2 = \lambda^* \circ i_2''$. Similarly, we get $\sigma_1 = \lambda^* \circ i_1''$, and we must have $\lambda^* = \lambda$.

So, $\operatorname{coker} \varphi \oplus \operatorname{coker} \psi$ satisfies the universal property for $\operatorname{coker}(\varphi \oplus \psi)$. $\square$

§5 Classification of finitely generated modules over PIDs

5.5

Recall (Exercise V.4.11) that a commutative ring is local if it has a single maximal ideal $\mathfrak{m}$. Let R be a local ring, and let M be a direct summand of a finitely generated free R-module: that is, there exists an R-module N such that $M \oplus N$ is a free R-module.

- Choose elements $m_1, \ldots, m_r \in M$ whose cosets mod $\mathfrak{m}M$ are a basis of $M/\mathfrak{m}M$ as a vector space over the field $R/\mathfrak{m}$. By Nakayama's lemma, $M = \langle m_1, \ldots, m_r \rangle$ (Exercise 3.10).
- Obtain a surjective homomorphism $\pi : F = R^{\oplus r} \to M$.
- Show that π splits, giving an isomorphism $F \simeq M \oplus \ker \pi$. (Apply Exercise III.6.9 to the surjective homomorphism π and the free module $M \oplus N$ to obtain a splitting $M \to F$; then use Proposition III.7.5.)
- Show $\ker \pi / \mathfrak{m} \ker \pi = 0$. Use Nakayama's lemma (Exercise 3.8) to deduce that $\ker \pi = 0$.
- Conclude that $M \simeq F$ is in fact free.

Solution.
First point is already proved in 3.10.

Second point is obvious.

By III.6.9, there is a morphism $\varphi : M \oplus N \to F$ such that the diagram

$$\begin{tikzcd} & M \oplus N \arrow[dl, "\varphi"'] \arrow[dr, "\pi_1"] & \\ F \arrow[rr, "\pi"'] & & M \end{tikzcd}$$

commutes. Define $\tau : M \to F$ by $\tau = \varphi \circ i_1$. We have $\pi \circ \tau = \pi \circ \varphi \circ i_1 = \pi_1 \circ i_1 = \mathrm{id}$, so τ is right-inverse for π and $F = M \oplus \ker \pi$.

Now I claim that
$$F/\mathfrak{m}F \simeq M/\mathfrak{m}M \oplus \ker \pi / \mathfrak{m} \ker \pi$$
as $R/\mathfrak{m}$-vector spaces and
$$\dim F/\mathfrak{m}F = \dim M/\mathfrak{m}M = r$$
It follows then that $\ker \pi / \mathfrak{m} \ker \pi = 0$, as desired.

To verify my claim, consider an R-module homomorphism

$$\psi : F \to M/\mathfrak{m}M \oplus \ker \pi / \mathfrak{m} \ker \pi$$

$$(m, k) \mapsto (m + \mathfrak{m}M, k + \mathfrak{m} \ker \pi)$$

If $\psi(m,k)=0$, then $m \in \mathfrak{m}M$, $k \in \mathfrak{m}\ker\pi$, hence for some $r_i, r_i' \in \mathfrak{m}$, $m_i' \in M$, $k_i \in \ker\pi$,

$$(m,k) = \left(\sum r_i' m_i', \sum r_i k_i\right) = \sum r_i'(m_i', 0) + \sum r_i(0, k_i) \in \mathfrak{m}F$$

On the other hand, if $f \in \mathfrak{m}F$, then

$$f = \sum r_i f_i = \sum r_i(m_i', k_i) = \left(\sum r_i m_i', \sum r_i k_i\right)$$

and $\psi(f)=0$. Since ψ is clearly surjective, we have an induced isomorphism of R-modules

$$\psi^* : F/\mathfrak{m}F \to M/\mathfrak{m}M \oplus \ker\pi/\mathfrak{m}\ker\pi$$

Moreover, ψ^* is a homomorphism of $R/\mathfrak{m}$-vector spaces. Indeed

$$\psi^*((r+\mathfrak{m})\bar{f}) = \psi^*(r\bar{f}) = r\psi^*(\bar{f}) = (r+\mathfrak{m})\psi^*(\bar{f})$$

and ψ^* also preserves sum as an R-module homomorphism. Consequently,

$$F/\mathfrak{m}F \simeq M/\mathfrak{m}M \oplus \ker\pi/\mathfrak{m}\ker\pi$$

as $R/\mathfrak{m}$ vector spaces.

Let $e_1, \ldots, e_r$ be a basis of F. The corresponding classes generate $F/\mathfrak{m}F$ over $R/\mathfrak{m}$ and if $\sum(r_i+\mathfrak{m})\bar{e}_i = \bar{0}$, then $\sum r_i e_i \in \mathfrak{m}F$, which implies $r_i \in \mathfrak{m}$ and $r_i + \mathfrak{m} = 0 + \mathfrak{m}$. Hence, $\bar{e}_1, \ldots, \bar{e}_r$ are linearly independent in $F/\mathfrak{m}F$ over $R/\mathfrak{m}$ and $\dim F/\mathfrak{m}F = r$.

As discussed above, we have now $\ker\pi/\mathfrak{m}\ker\pi = 0$ as $R/\mathfrak{m}$-vector space, and hence as an R-module. It implies that $\ker\pi = \mathfrak{m}\ker\pi$ and $\ker\pi = 0$ by Nakayama's lemma ($\ker\pi \simeq F/M$ is finitely generated).

Thus, π is an isomorphism and $M \simeq R^r$. □

5.6

Let R be an integral domain, and let $M = \langle m_1, \ldots, m_r\rangle$ be a finitely generated module. Prove that $\operatorname{rk} M \leq r$.

Solution.
Since M is generated by r elements, there is a surjective homomorphism $R^r \to M$, meaning $M \simeq R^r/N$ for some submodule N. Every linear relation in R^r reduces modulo N to a relation in M, so there can't be more than r linearly independent elements in M. □

5.7

Let R be an integral domain, and let M be a finitely generated module over R. Prove that $\operatorname{rk} M = \operatorname{rk}(M/\operatorname{Tor}(M))$.

Solution.
If a system of elements in M is linearly dependent, then its image in $M/\operatorname{Tor}(M)$ under canonical projection is also linearly dependent. Hence, $\operatorname{rk} M \geq \operatorname{rk}(M/\operatorname{Tor}(M))$.

For the reverse direction, consider $\overline{m_1}, \dots, \overline{m_r} \in M/\operatorname{Tor}(M)$ such that

$$\sum r_i \overline{m_i} = \overline{0}$$

It follows that $\sum r_i m_i \in \operatorname{Tor}(M)$, so there is some $s \in R \setminus \{0\}$ such that $\sum s r_i m_i = 0$, meaning that $\{m_1, \dots, m_r\}$ is linearly dependent in M. Hence, $\operatorname{rk} M \leq \operatorname{rk}(M/\operatorname{Tor}(M))$. $\square$

5.8

Let R be an integral domain, and let M be a finitely generated module over R. Prove that $\operatorname{rk} M = r$ if and only if M has a free submodule $N \simeq R^r$, such that M/N is torsion.

If R is a PID, then N may be chosen so that $0 \to N \to M \to M/N \to 0$ splits.

Solution.
$\Longrightarrow$ Suppose $m_1, \dots, m_r$ are linearly independent. They generate

$$N = \langle m_1, \dots, m_r \rangle \simeq R^r$$

If $\overline{m} \in M/N$, $\overline{m} \neq \overline{0}$, then $m_1, \dots, m_r, m$ are linearly dependent $\implies \exists r \neq 0, r_i$ not all zeros, such that $\sum r_i m_i + rm = 0$. $\implies r\overline{m} = \overline{-\sum r_i m_i} = \overline{0}$, so M/N is torsion.

$\Longleftarrow$ Let N be a submodule of M such that $N \simeq R^r$, M/N is torsion, and let $n_1, \dots, n_r$ be its basis.

$n_1, \dots, n_r$ is a maximal linearly independent system of M. Indeed, if $m \in M$, then $\exists r \in R$ such that $rm \in N$, since M/N is torsion. $rm = \sum r_i n_i$, so $n_1, \dots, n_r, m$ is linearly dependent.

Now if $m_1, \dots, m_s$ is linearly independent, then $s \leq r$, so $\operatorname{rk} M = r$.

Suppose now R is a PID and let K be a finitely generated torsion-free R-module. Let us show that K is free. K has a free submodule $F \simeq R^s$, such that K/F is torsion. By 4.4, $\exists a \in \operatorname{Ann}(K/F)$ such that $a \neq 0$. The morphism $\varphi : K \to F$, $k \mapsto ak$ is injective, since K is torsion-free, and identifies K with a submodule of F

$$K \simeq \operatorname{im} \varphi \subset F$$

Since over a PID every submodule of a free module is free, we conclude that K is free.

If M is a finitely generated module over a PID R with $\operatorname{rk} M = r$, then $M/\operatorname{Tor}(M)$ is torsion-free, hence free, and by 5.7

$$\operatorname{rk}(M/\operatorname{Tor}(M)) = \operatorname{rk} M = r \implies M/\operatorname{Tor}(M) \simeq R^r$$

Let $m_1, \dots, m_r \in M$ be elements whose classes form a basis of $M/\operatorname{Tor}(M)$ and consider

$$N = \langle m_1, \dots, m_r \rangle \simeq R^r$$

It follows that $N + \operatorname{Tor}(M) = M \implies M \simeq N \oplus \operatorname{Tor}(M)$, since $N \cap \operatorname{Tor}(M) = 0$.

Thus, $M/N \simeq \operatorname{Tor}(M)$ is torsion and

$$0 \to N \to M \to M/N \to 0$$

splits. □

5.9

Let R be an integral domain, and let

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of finitely generated R-modules. Prove that $\operatorname{rk} M_2 = \operatorname{rk} M_1 + \operatorname{rk} M_3$.

Deduce that 'rank' defines a homomorphism from the Grothendieck group of the category of finitely generated R-modules to $\mathbb{Z}$

Solution.
Denote $\operatorname{rk} M_i = r_i$ and suppose that $\{m_1, \dots, m_{r_1}\}$ and $\{m''_1, \dots, m''_{r_3}\}$ are linearly independent systems in M_1 and M_3 respectively. Let us denote the morphisms in our sequence by λ and λ'.

Consider elements $m'_1, \dots, m'_{r_3} \in M_2$ such that $\lambda'(m'_k) = m''_k$. Then the system

$$\lambda(m_1), \dots, \lambda(m_{r_1}), m'_1, \dots, m'_{r_3}$$

is linearly independent. Indeed, if $\sum s_k \lambda(m_k) + \sum s'_k m'_k = 0$, then

$$0 = \lambda' \left(\sum s_k \lambda(m_k) + \sum s'_k m'_k \right) = \sum s'_k \lambda'(m'_k) = \sum s'_k m''_k \implies s'_k = 0,$$

since $m''_1, \dots, m''_{r_3}$ is linearly independent. We have now

$$\lambda \left(\sum s_k m_k \right) = \sum s_k \lambda(m_k) = 0 \implies \sum s_k m_k = 0,$$

since λ is an inclusion $\implies s_k = 0$.

Now if $m' \in M_2$, then there exist $s'' \neq 0$ and s''_k for $1 \leq k \leq r_3$, such that

$$s'' \lambda'(m') + \sum s''_k \lambda'(m'_k) = 0$$

Since $\ker \lambda' = \operatorname{im} \lambda$, it follows that there exists $m \in M_1$ such that

$$\lambda(m) = s'' m' + \sum s''_k m'_k$$

On the other hand, there exist $s \neq 0$ and s_k for $1 \leq k \leq r_1$, such that $sm + \sum s_k m_k = 0$. We have then

$$s s'' m' + \sum s s''_k m'_k + \sum s_k \lambda(m_k) = 0$$

Because R is an integral domain and $s \neq 0, s'' \neq 0$, their product $ss'' \neq 0$. Thus,

$$\lambda(m_1), \ldots, \lambda(m_{r_1}), m'_1, \ldots, m'_{r_3}$$

is a maximal linearly independent system in M_2, and hence every other linearly independent system in M_2 has $\leq r_1 + r_3$ elements, which concludes the proof. □

5.13

Let M be a finitely generated module over an integral domain R. Prove that if R is a PID, then M is torsion-free if and only if it is free. Prove that this property characterizes PIDs. (Cf. Exercise 4.3.)

Solution.
If M is torsion-free, then it is free as shown in 5.8.

If M is free with basis $b_1, \ldots, b_r$ and has a torsion m, so that $rm = 0$, $r \neq 0$, then

$$0 = rm = r \sum r_i b_i = \sum r r_i b_i \implies r r_i = 0 \ \forall i$$

$$\implies r_i = 0 \ \forall i \quad \text{and} \quad m = 0,$$

so M is torsion-free.

Now we want to show that if a ring R has a property that for any finitely generated R-module M,

$$M \text{ is torsion-free} \iff M \text{ is free}$$

Then R is a PID.

Indeed, R is free as an R-module: $\{1\}$ is a linearly independent system, for if $r \cdot 1 = 0$, then $r = 0$; and $R = (1)$. Consequently, R is torsion-free, so it is an integral domain.

Consider an ideal $I \subset R$. Since R is an integral domain, I is torsion-free, hence free with $\operatorname{rk} I \leq 1$. It follows that $I = (a)$ for some a. □